Prince Edward Island
CANADA

Education and Early Childhood Development

Specific Curriculum Outcomes Framework

Grade 4

2013

Table of Contents

Introduction iii

Communication and Information Technology 1

Core French 5

English Language Arts 7

Health 10

Mathematics 12

Music 14

Physical Education 15

Science 18

Social Studies 23

Visual Arts 24

Introduction

This learning outcomes framework comprises a series of specific curriculum outcome statements describing what knowledge, skills, and attitudes students are expected to demonstrate as a result of their cumulative learning experiences in their public school education. Through an ongoing process, the Department of Education and Early Childhood Development is developing a learning outcomes framework for each area of the public school program.

This document provides an overview of the learning outcomes framework organized by grade level and subject area. It is intended to serve as a brief survey of expected learning outcomes and as a tool to assist teachers in program planning. The connections among learning outcomes reflect natural affinities among subject areas and facilitate the design of a balanced, integrated program.

In designing appropriate learning experiences that enable students to achieve the expected learning outcomes, teachers and administrators are expected to refer to foundation documents and related curriculum guides. In planning the appropriate use of information technologies as tools for learning and teaching, teachers and administrators should also refer to the *Journey On* documents, located on the Department of Education and Early Childhood Development website http://www.edu.pe.ca/journeyon/

Foundation documents provide the framework for general and key-stage curriculum outcomes, outline the focus and key features of the curriculum, and describe contexts for learning and teaching. Curriculum guides elaborate on specific curriculum outcomes and describe other aspects of curriculum, such as program design and components, instructional and assessment strategies, and resources.

General curriculum outcomes are statements which identify what students are expected to know and be able to do upon completion of study in a curriculum area. Key-stage curriculum outcomes are statements which identify what students are expected to know and be able to do by the end of Grades 3, 6, 9, and 12 as a result of their cumulative learning experiences in a curriculum area. Specific curriculum outcomes are statements which identify what students are expected to know and be able to do at a particular grade level.

The following overview of the learning outcomes framework notes general curriculum outcomes and specific curriculum outcomes.

Elementary Program Components

Elementary schools must include, for all children in each year's program for kindergarten through Grades 1-6, health, language arts, mathematics, music, physical education, social studies, science, and visual arts. The elementary program does not include technology education as a subject area; however, the general and key-stage curriculum outcomes for technology education included in this booklet provide a framework for teachers of kindergarten through Grades 1-6 to use in integrating technology education within learning experiences across the curriculum.

Communication and Information Technology

General Technology Outcomes

GTO A - Technology Problem Solving
Students will be expected to design, develop, evaluate, and articulate technological solutions

GTO B - Technology Systems
Students will be expected to operate and manage technological systems

GTO C - History and Evolution of Technology
Students will be expected to demonstrate an understanding of the history and evolution of technology and of its social and cultural implications

GTO D - Technology and Careers
Students will be expected to demonstrate an understanding of current and evolving careers and of the influence of technology on the nature of work

GTO E - Technological Responsibility
Students will be expected to demonstrate an understanding of the consequences of their technological choices

CODES USED IN CONTINUUM

(A) AWARENESS LEVEL
The student is exposed to the technology as it is being used by others

(G) GUIDED LEVEL
The student begins to use the technology with the help of others

(I) INDEPENDENT LEVEL
The student uses technology without assistance

Specific Curriculum Outcomes

Computer System

Students will be expected to

A1.1 (A) make use of help features to independently find solutions to problems
B1.1 (I) login, open, and close a program; open, save, and manage technological solutions; close a file with a mouse
B1.2 (G) demonstrate proper use of login numbers and names, set-up and change passwords, and be aware of the implications of multiple logins
B1.3 (G) begin to work with more than one file open at once (multitask)
B1.4 (I) differentiate between “Save” and “Save as”
B1.5 (A) be able to identify the common Windows components of a given software screen (menu bar, button bar, cursor, insertion point)
B1.6 (A) have an understanding of file management (drives and folders, rename, select, move, copy, paste, delete, display format, backup, etc.)
B1.8 (A) understand the difference between software and hardware
B1.10 (G) understand how and when to re-boot (warm boot vs cold boot)
B1.12 (G) demonstrate proper use of network printing, choosing printer, recognizing process and purpose of print queues
B1.13 (A) identify computer viruses, how they are transmitted, and how anti-virus software is used to protect or clean a computer
B1.14 (A) identify SPAM, pop-up ads, spyware, and other invasive software coding
B1.15 (A) modify and utilize pages/templates
C1.1 (I) identify technologies that are found in everyday life

Social, Ethical and Health

Students will be expected to

A2.1 (G) identify aspects of an ergonomic workstation (lighting, monitor angle, work placement, keyboard height, seat height, posture, etc.)
B2.1 (G) demonstrate proper touch keyboarding techniques (i.e., home row, quick key strokes, proper reaches)
D2.1 (A) determine the technological requirements for specific career goals
E2.1 (I) respect equipment and other students’ work
E2.2 (I) work cooperatively at work station
E2.3 (I) adhere to Acceptable Use Agreement for work stations/network/Internet
E2.4 (G) use electronic communication etiquette
E2.7 (A) identify ethical issues involved with Internet content, awareness of inappropriate use of technology
E2.8 (A) demonstrate caution before sending personal information over the Internet
E2.9 (A) follow publishing etiquette (suitable language, no discrimination, etc.) adhere to the guidelines for school web pages as outlined by PEI Department of Education and Early Childhood Development

Internet

Students will be expected to

A3.1 (G) demonstrate awareness of the Internet as a source of information

General Technology Outcomes

Specific Curriculum Outcomes

Students will be expected to

A3.2 (A) use various tools (search engines and directories) and strategies necessary to carry out research
A3.3 (A) obtain/download material (test, graphics, files) from the Internet
B3.1 (G) use the various browser navigation tools (back, forward, history)
B3.2 (G) manage bookmarks/favourites
B3.3 (A) distinguish among various file formats (file extensions), required plugins, file compression/decompression utilities
C3.1 (A) discuss ways in which the Internet is evolving
E3.1 (A) critically evaluate information and its source based on pre-determined criteria

Concept Maps

Students will be expected to

A4.1 (G) use brainstorming techniques to generate ideas
A4.2 (A) create a web (i.e., literary, concept, character, word, Venn Diagrams, and timelines)
A4.3 (I) categorize ideas graphically
A4.4 (I) create links between ideas; re-link or delete links between ideas
B4.1 (G) add fonts, graphics, sound and colours to enhance ideas
A4.5 (I) elaborate on ideas (i.e., adding notes, annotations, etc.)
B4.2 (A) create hyperlinks to files, web sites, or multimedia content

Graphics

Students will be expected to

A5.1 (G) create illustrations or graphics by using the various drawing tools
A5.2 (A) apply principles of design
B5.1 (G) demonstrate various object editing features (i.e., select, unselect, resize, crop, area fill, add colour and pattern, size adjustment using the mouse or scale, various erasing techniques, object orientation, change font and text size, colour or appearance, create text blocks, change text wrap selection, and other text manipulation functions)
B5.2 (A) carry out various object manipulations (i.e., object alignment, creation of graphics in layers, grouping/un-grouping components of an image

Spreadsheets

Students will be expected to

A6.1 (A) plan/design a spreadsheet to organize and tabulate data from various sources to make a schedule, tally/score sheet, solve a mathematical word problem
A6.2 (A) correct errors, modify, or delete data in a cell
A6.4 (A) use different types of graphs/charts (line, pie, bar) to visually represent data; label graph components (legend, title, *x-y* axis, colour, fill pattern)
B6.1 (A) identify spreadsheet components and terminology (rows and columns, cell addresses, data entry bar)
B6.3 (A) enter data into simple pre-existing spreadsheets, auto fill data, data entry bar, sort data

General Curriculum Outcomes

Specific Curriculum Outcomes

Students will be expected to

B6.4 (A) edit spreadsheet layout (insert and delete rows or columns, select a range of cells, alter column widths and row heights, lock row and column headings, lock and unlock cell(s), create fixed titles)

B6.6 (A) format numbers (decimal places, currency, etc.), format text (font, colour, size)

Word Processing

Students will be expected to

A7.2 (I) identify examples of desktop publishing (i.e., newspaper, catalogue, ads, brochure)

B7.1 (I) use a grade level appropriate word processor to create and edit written work

B7.2 (I) locate characters on a keyboard and identify functions of word processing (i.e., cursor, insertion point, enter key, space bar, upper case, backspace, shortcut key)

B7.3 (G) use editing tools to revise work (i.e., spell check, thesaurus, find, and replace)

B7.4 (I) change font, size, colour, style (i.e., bold italics, underline, insert special characters, drop capitals)

B7.5 (G) format text (i.e., justification, line spacing, outlines and bullets, text wrap)

B7.6 (A) format documents (i.e., using margins, tab rulers, indents, page centre, border, watermark)

B7.7 (G) insert a graphic and manipulate (i.e., resize, add borders and fill, create text art)

B7.8 (A) insert and format tables and text boxes (i.e., lines, fill, columns, rows, borders, alignment)

Multimedia

Students will be expected to

A8.1 (A) apply planning strategies (storyboards, scripts, graphic organizing, brainstorming)

A8.2 (A) create an age/grade appropriate slide show presentation that may contain one or more of the following objects (text, graphics, images, animations, audio, and video)

A8.5 (A) select an appropriate medium to convey message (be conscious of file size, formats, and storage location)

B8.1 (I) navigate multimedia resources such as slide shows, on-line resources or CD-ROM interactive educational activities

Database

Students will be expected to

A9.1 (I) use an existing database (CD-ROM, MicroCat, Dynex, Internet search engine) to find information (sign up for Provincial Library Card - Abbycat)

A9.2 (G) perform searches on a database file using logical and Boolean operators (understands commands, scope, filters, and conditions)

A9.3 (A) design/plan a database to use as a method of organizing information

A9.4 (A) create and modify a form (add graphics and error checking routines)

General Technology Outcomes

Specific Curriculum Outcomes

Students will be expected to

A9.5 (A) use databases to analyse data and look for trends
B9.1 (G) enter data into a pre-existing database, edit data, and use automated text
B9.2 (A) create fields with variable field types (numeric, text, date) and properties (colour, width, font, etc.)
B9.3 (A) restructure database (add/delete fields, change field width)
B9.4 (A) sort records alphabetically, numerically, and by multiple fields
B9.5 (A) create a report from the entire database or selected records
E9.1 (A) examine functions and implications of database driven web sites (i.e., on-line purchasing, searching, and password secure items)

Telecommunications

Email:

Students will be expected to

B10.1 (I) send messages
B10.2 (I) open messages
B10.3 (A) manage mail/folders
B10.4 (A) manage address books
B10.5 (A) use distribution lists
B10.6 (A) send and open attachments
B10.7 (A) create signatures

E-Learning Collaborative Tools:

Students will be expected to

A10.1 (A) collaborate using software (i.e., whiteboard, slideshow, application sharing, chat, messaging, send and receive files, photos, group file sharing, resource sharing [links], on-line content creation and sharing, assignment drop box, video and audio, discussion forums, journal)

Web Authoring

Students will be expected to

A11.1 (G) identify web page creation possibilities

Core French

General Curriculum Outcomes

Communication

GCO 1: On the basis of their experience in the core French curriculum, students should be able to communicate effectively in French, both orally and in writing, and to interact appropriately in a variety of situations that relate to their needs and interests.

General Language Education

GCO 2: On the basis of their experience in the core French curriculum, students should be able to choose and implement strategies to facilitate their communication in French and improve their learning.

Culture

GCO 3: On the basis of their experience in the core French curriculum, students should be able to demonstrate an appreciation and understanding of Francophone cultures, while comparing them with their own culture, as well as an appreciation and understanding of Canada's multicultural reality.

Key-Stage Curriculum Outcomes and Sample Specific Curriculum Outcomes

(For a complete list of the specific curriculum outcomes, please refer to the Elementary Core French Curriculum Guide.)

Students should be able to

1.1 cope in a classroom where French is the language spoken (e.g., follow the instructions of the teacher, ask permission)
1.2 share personal information by using partial sentences and/or by answering questions (e.g., name, age, date of birth)
1.3 identify and describe objects, animals, people, events, and places that are part of their environment (e.g., holidays, family members, colours)
1.4 share their tastes, preferences, interests, and feelings (e.g., favourite animal, favourite season, feelings)
1.5 participate in a variety of interactive activities (e.g., play games, interview a partner)
1.6 ask simple questions (e.g., Est-ce que je peux...?)
1.7 demonstrate an understanding of the main idea in a simple text (e.g., answer global comprehension questions)
1.8 select relevant details in a simple text (e.g., important words)
1.9 respond personally to simple texts, often by following a model (e.g., mime, drawing)
1.10 produce a variety of simple texts, often by following a model (e.g., card, poster, invitation)

2.1 use various clues to recognize and identify words, for example, words in the same family, prefixes, suffixes, cognates, and loanwords
2.2 use textual clues to anticipate and understand a text, for example, pictures, graphic representations, page layout, listening to key words and context (e.g., graphic organizers, illustrations)
2.3 create connections between their prior knowledge and the text being studied (e.g., make predictions)
2.4 recognize the importance of their role in the learning process, their responsibilities toward the learning process and their responsibilities toward others, for example, by cooperation, interaction, reflecting on what is being taught, risk taking, etc. (e.g., take risks, participate actively)
2.5 use a variety of tools, for example, dictionaries, and technology to create text
2.6 produce an oral or written text by following a production model, for example, the writing process

3.1 identify and locate certain francophone communities locally, provincially, nationally, and internationally (e.g., realize French schools exist in our province)
3.2 identify several contemporary francophone personalities (e.g., Céline Dion)
3.3 demonstrate an awareness of Canada's cultural mosaic (e.g., demonstrate a respect towards other cultures)
3.4 demonstrate knowledge of authentic texts (e.g., sing songs in French)

General Curriculum Outcomes

Key-Stage Curriculum Outcomes and Sample Specific Curriculum Outcomes

Students should be able to

Language

GCO 4: On the basis of their experience in the core French curriculum, students should be able to recognize and use in context elements of the linguistic code, orally and in writing, to facilitate their communication in French.

4.1 recognize the linguistic elements relating to areas of experience and to their communication needs (e.g., understand, orally and in writing, the vocabulary associated with the themes)

4.2 use the linguistic elements relating to areas of experience and their communication needs (e.g., use singular pronouns in context)

Fields of Experience

GCO 5: On the basis of their experience in the core French curriculum, students should be able to participate in a variety of language experiences appropriate to their needs and interests.

5.1 engage in language experiences in a variety of areas (e.g., family, animals, clothing)

English Language Arts

General Curriculum Outcomes	Specific Curriculum Outcomes	
	Students will be expected to	
Speaking and Listening	1.1	explore and discuss their thoughts, ideas, and experiences and consider those of their peers
GCO 1: Students will be expected to speak and listen to explore, extend, clarify, and reflect on their thoughts, ideas, feelings, and experiences.	1.2	ask and respond to questions to clarify information and explore solutions to problems (e.g., using an interview format)
	1.3	explain personal opinions and respond to the questions and opinions of others
	1.4	listen critically to others' ideas or opinions expressed
GCO 2: Students will be expected to communicate information and ideas effectively and clearly, and to respond personally and critically.	2.1	contribute to conversations, small-group and whole-group discussion, showing an awareness of when to speak and when to listen
	2.2	use word choice, tone of voice, facial expressions, and gestures appropriate to the speaking occasion
	2.3	give and follow instructions and respond to questions and directions
	2.4	engage in and respond to oral presentations (e.g., retell a story, sing a song)
GCO 3: Students will be expected to interact with sensitivity and respect, considering the situation, audience, and purpose.	3.1	show basic courtesies of conversation in group interactions
	3.2	identify examples of prejudice and stereotyping in oral language, and use language that shows respect for all people
	3.3	show an awareness of the kinds of language appropriate to different situations and audiences
Reading and Viewing	4.1	select, with growing independence, texts appropriate to their interests and learning needs
GCO 4: Students will be expected to select, read, and view with understanding a range of literature, information, media, and visual texts.	4.2	read widely and experience a variety of children's literature with an emphasis in genre and authors
	4.3	use pictures and illustrations, word structures, and text features (e.g., table of contents, headings and subheadings, glossaries, structures of narrative and expository texts, key ideas, and margin notes) to locate topics and obtain or verify understandings of information
	4.4	use and integrate the pragmatic, semantic, syntactic, and graphophonic cueing systems (including context clues, word order, suffixes, compound words, contractions, and singular and plural words) and a variety of strategies to construct meaning
	4.5	describe their own processes and strategies in reading and viewing
GCO 5: Students will be expected to interpret, select, and combine information using a variety of strategies, resources, and technologies.	5.1	answer, with assistance, their own and others' questions by seeking information from a variety of texts - determine their own and community (class) needs for information - recognize the purpose of classification systems and basic reference materials

General Curriculum Outcomes	Specific Curriculum Outcomes *Students will be expected to*	
		- use a range of reference texts and a database or an electronic search to facilitate the selection process - reflect on the process of generating and responding to their own and others' questions
GCO 6: Students will be expected to respond personally to a range of texts.	6.1	describe, share, and discuss their personal reactions to texts
	6.2	give reasons for their opinions about texts and types of texts, and the work of authors and illustrators
GCO 7: Students will be expected to respond critically to a range of texts, applying their understanding of language, form, and genre.	7.1	use their background knowledge to question information presented in print and visual texts
	7.2	identify conventions and characteristics of different types of print and media texts that help them understand what they read and view
	7.3	respond critically to texts by - asking questions and formulating understandings - discussing texts from the perspective of their own experiences - identifying instances where language is being used, not only to entertain, but to manipulate, persuade, or control them - identifying instances of prejudice and stereotyping
GCO 8: Students will be expected to use writing and other forms of representation to explore, clarify, and reflect on their thoughts, feelings, experiences, and learnings; and to use their imaginations.	8.1	use strategies in writing and other ways of representing to - formulate questions and organize ideas - generate topics of personal interest and importance - discover and express personal attitudes, feelings, and opinions - compare their own thoughts and beliefs to those of others - describe feelings, reactions, values, and attitudes - record experiences - formulate goals for learning - practise strategies for monitoring their own learning
	8.2	experiment with different ways of making their own notes (e.g., webbing, jot notes, matrix)
	8.3	experiment with language appropriate to purpose, audience, and form, that enhances meaning and demonstrates imagination in writing and other ways of representing
GCO 9: Students will be expected to create texts collaboratively and independently, using a wide variety of forms for a range of audiences and purposes.	9.1	create written and media texts, collaboratively and independently, in different modes (expressive, transactional, and poetic) and in a variety of forms - recognize that particular forms require the use of specific features, structures, and patterns
	9.2	demonstrate an awareness of purpose and audience
	9.3	invite responses to early drafts of their writing/media productions - use audience reaction to help shape subsequent drafts

General Curriculum Outcomes

GCO 10: Students will be expected to use a range of strategies to develop effective writing and other ways of representing and to enhance their clarity, precision, and effectiveness.

Specific Curriculum Outcomes

Students will be expected to

10.1 develop a range of prewriting, drafting, revising, editing, proofreading, and presentation strategies

10.2 demonstrate an understanding of many conventions of written language in final products
- correctly spell many familiar and commonly used words
- demonstrate an increasing understanding of punctuation, capitalization, and paragraphing
- demonstrate a growing awareness of appropriate syntax
- use references while editing (e.g., dictionaries, classroom charts, electronic spell checkers, checklists)

10.3 use technology with increasing proficiency in writing and other forms of representing

10.4 demonstrate a commitment to shaping pieces of writing and other representations through stages of development

10.5 select, organize, and combine relevant information from two or more sources to construct and communicate meaning

Health

General Curriculum Outcomes

Wellness Choices

GCO: Students will make responsible and informed choices to maintain health and to promote safety for self and others.

Relationship Choices

GCO: Students will develop effective interpersonal skills that demonstrate responsibility, respect and caring in order to establish and maintain healthy interactions.

Specific Curriculum Outcomes

Personal Health

Students will be expected to

W-4.1 demonstrate an understanding of the connections among physical activity, emotional wellness, and social wellness
W-4.2 evaluate the impact of environmental factors on personal health, and develop positive environmental health habits
W-4.3 examine the various factors that influence body image
W-4.4 analyse the need for variety and moderation in a balanced diet
W-4.5 examine and evaluate the health risks associated with smoking and other forms of tobacco use

Safety and Responsibility

Students will be expected to

W-4.6 describe and demonstrate an understanding of passive, aggressive, and assertive behaviours
W-4.7 expand practices that provide safety for self and others
W-4.8 describe ways to respond appropriately to potentially dangerous situations related to environmental conditions
W-4.9 describe and demonstrate ways to assist with the safety of others

Understanding and Expressing Feelings

Students will be expected to

R-4.1 recognize that individuals can have a positive and negative influence on the feelings of others
R-4.2 identify and use long-term strategies for managing anger
R-4.3 recognize that management of positive/negative stress can affect health
R-4.4 describe and demonstrate communication skills and behaviours that show respect for the feelings of others

Interactions

Students will be expected to

R-4.5 identify changes that occur in friendships, and explore strategies to deal with changes
R-4.6 identify and describe ways to provide support to others
R-4.7 demonstrate an understanding of effective communication skills and behaviours to reduce escalation of conflict

Group Roles and Processes

Students will be expected to

R-4.8 describe roles and responsibilities within a group
R-4.9 assess how to act as an important role model to others

General Curriculum Outcomes

Life Learning Choices

GCO: Students will use resources effectively to manage and explore life roles and career opportunities and challenges.

Specific Curriculum Outcomes

Students will be expected to

Learning Strategies

Students will be expected to

L-4.1 develop and apply skills for personal organization/study
L-4.2 identify ways individuals continue to learn throughout their lives
L-4.3 demonstrate effective decision making
L-4.4 distinguish between and set different kinds of goals

Life Goals and Career Development

Students will be expected to

L-4.5 relate personal interests to various occupations
L-4.6 recognize that personal roles will change over time and circumstances

Volunteerism

Students will be expected to

L-4.7 describe the impact of service contributions on self
L-4.8 select, perform as a class, and analyse volunteer accomplishments

Mathematics

General Curriculum Outcomes

Number (N)
GCO: Develop number sense.

Specific Curriculum Outcomes

Students will be expected to

N1 Represent and describe whole numbers to 10 000, pictorially and symbolically.

N2 Compare and order numbers to 10 000.

N3 Demonstrate an understanding of addition of numbers with answers to 10 000 and their corresponding subtractions (limited to 3- and 4-digit numerals) by:
- using personal strategies for adding and subtracting;
- estimating sums and differences;
- solving problems involving addition and subtraction.

N4 Explain the properties of 0 and 1 for multiplication and the property of 1 for division.

N5 Describe and apply mental mathematics strategies, such as:
- skip counting from a known fact;
- using doubling or halving;
- using doubling or halving and adding or subtracting one more group;
- using patterns in the 9s facts;
- determine basic multiplication facts to 9x9 and related division facts.

N6 Demonstrate an understanding of multiplication (2- or 3-digit by 1-digit) to solve problems by:
- using personal strategies for multiplication with and without concrete materials;
- using arrays to represent multiplications;
- connecting concrete representations to symbolic representations;
- estimating products.

N7 Demonstrate an understanding of division (1-digit divisor and up to 2-digit dividend) to solve problems by:
- using personal strategies for dividing with and without concrete materials;
- estimating quotients;
- relating division to multiplication.

N8 Demonstrate an understanding of fractions less than or equal to one by using concrete and pictorial representations to:
- name and record fractions for the parts of a whole or a set;
- compare and order fractions;
- model and explain that for different wholes, two identical fractions may not represent the same quantity;
- provide examples of where fractions are used.

N9 Describe and represent decimals (tenths and hundredths) concretely, pictorically, and symbolically.

N10 Relate decimals to fractions (to hundredths).

N11 Demonstrte an understanding of addition and subtraction of decimals (limited to hundredths) by:
- using compatible numbers;
- estimating sums and differences;
- using mental math strategies to solve problems.

General Curriculum Outcomes	Specific Curriculum Outcomes	
	Students will be expected to	
Patterns and Relations (PR) GCO: Use patterns to describe the world and solve problems.	PR1	Identify and describe patterns found in tables and charts, including a multiplication chart.
	PR2	Reproduce a pattern shown in a table or chart using concrete materials.
	PR3	Represent and describe patterns and relationships using charts and tables to solve problems.
	PR4	Identify and explain mathematical relationships using charts and diagrams to solve problems.
	PR5	Express a given problem as an equation in which a symbol is used to represent an unknown number.
	PR6	Solve one-step equations involving a symbol to represent an unknown number.
Shape and Space (SS) GCO: Use direct and indirect measure to solve problems.	SS1	Read and record time using digital and analog clocks, including 24-hour clocks.
	SS2	Read and record calendar dates in a variety of formats.
	SS3	Demonstrate an understanding of area of regular and irregular 2-D shapes by: • recognizing that area is measured in square units; • selecting and justifying referents for the units cm^2 or m^2; • estimating area by using referents for cm^2 or m^2; • determining and recording area (cm^2 or m^2); • constructing different rectangles for a given area (cm^2 or m^2) in order to demonstrate that many different rectangles may have the same area.
	SS4	Describe and construct rectangular and triangular prisms.
	SS5	Demonstrate an understanding of line symmetry by: • identifying symmetrical 2-D shapes; • creating symmetrical 2-D shapes; • drawing one or more lines of symmetry in a 2-D shape.
	SS6	Demonstrate an understanding of congruency, concretely and pictorially.
Statistics and Probability (SP) GCO: Collect, display, and analyse data to solve problems.	SP1	Demonstrate an understanding of many-to-one correspondence.
	SP2	Construct and interpret pictographs and bar graphs involving many-to-one correspondence to draw conclusions.

Music

General Curriculum Outcomes

Musical Awareness and Appreciation

GCO 2: Students will be expected to develop an appreciation of the importance of music in daily life and respect the role that music plays in their heritage and culture.

Specific Curriculum Outcomes

Students will be expected to

4.2.1 sing alone and with others, with emphasis on expressive singing, phrasing, range, and more complex textures including two- and three-part rounds and canons

4.2.2 compare different and similar roles locally, nationally, and globally

4.2.3 compare (changes in) their thoughts and feeling with those of others about the role and influence of music in their daily lives including ways in which it is use in mass media and popular culture

4.2.4 demonstrate respect for the environmental, historical, and social factors that influence music of diverse cultures

4.2.5 discuss a respect for different and diverse musical cultures of the past and present

4.2.6 demonstrate their appreciation and understanding of artistic, kinesthetic, and social connections between music and the other arts

Physical Education

Please note: The three goals, Active Living, Skilful Movement, and Relationships will be referred throughout this section as GCO 1, GCO 2, and GCO 3.

GCO 1: Active Living
Enjoy and engage in healthy levels of participation in movement activities to support lifelong active living in the context of self, family, and community.

GCO 2: Skilful Movement
Enhance quality of movement by understanding, developing, and transferring movement concepts, skills, tactics, and strategies to a wide variety of movement activities.

GCO 3: Relationships
Balance self through safe and respectful personal, social, cultural, and environmental interactions in a wide variety of movement activites.

General Curriculum Outcomes

GCO 1

GCO 2

Note: Rolling is a safety skill that supports future participation in various movement activites. Grade 2 students need to be taught how to roll forward safely (see indicator 't'). For this Grade 2 outcome, teachers should ask students to show how they can roll backward and not challenge them to go beyond backward rolls of their own creation. Descriptors of how to roll backwards safely are provided here as they appear in the Grade 3 curriculum.

Specific Curriculum Outcomes

Active Living
Students will be expected to

4.1 Health-Related Fitness
Make decisions about and apply, with guidance, strategies and principles related to fitness improvement to determine own level of health-related fitness (cardiovascular endurance, flexibility, muscular endurance, and muscular strength) and to positively affect own level of health-related fitness.

Cross-curricular link
Health: Wellness Choices - W-4.1

4.2 Body Systems
Demonstrate an awareness of the body systems (circulatory, respiratory, and muscular) that are directly related to and affected by the development of the health-related components of fitness (cardiovascular endurance, muscular endurance, muscular strength, flexibility, and body composition)

Curriculum Note:
Body systems are taught in Grade 5 Science.

Skilful Movement
Students will be expected to

4.3 Complex Locomotor Skills
Select and apply performance cues to refine and combine locomotor skills into increasingly complex movement skills as applicable to lead-up games and body management activities, including dance and educational gymnastics, and others such as yoga, skipping, aerobics, martial arts, and track and field

4.4 Locomotor Skills
Apply, with guidance, how to skilfully perform locomotor skills while participating in movement activities, including at a
- utilization level of skill when
  - rolling backward (see note at the left)

General Curriculum Outcomes

Specific Curriculum Outcomes

Students will be expected to

4.5 Complex Non-Locomotor Skills
Select and apply performance cues to combine and refine non-locomotor skills of
- balancing
- jumping and landing on the spot on feet and hands
- rotating on the spot into increasingly complex movement skills while participating in body management activities (including dance and educational gymnastics, as well as others such as yoga, skipping, aerobics, martial arts, and track and field)

4.6 Manipulative Skills
Explore, express, and apply, with guidance, a variety of ways to skillfully move objects while participating in movement activities, including at a
- utilization level of skill when
  - hand dribbling
  - foot dribbling
  - striking objects with hands and/or short-handled implements (racquets and paddles)
- control level of skill when
  - volleying (to send an object in the air before it comes to rest)
  - striking objects with long-handled implements (bats, golf clubs, hockey sticks)
- progressing-towards-control level of skill when
  - punting

4.7 Complex Manipulative Skills
Select and apply performance cues to combine and refine manipulative (sending, receiving, and accompanying objects) skills in increasingly complex movement activities such lead-up games, including
- throwing
- catching (gathering, collecting)
- kicking

GCO 2, GCO 3

Skilful Movement, Relationships
Students will be expected to

4.8 Movement Refinement
Refine the application of movement variables, movement concepts, and performance cues to improve personal performance and to provide feedback to others

General Curriculum Outcomes	Specific Curriculum Outcomes
GCO 1, GCO 2, GCO 3	**Active Living, Skilful Movement, Relationships** *Students will be expected to* 4.9 Skilful Play Select and use effective movement skills, tactics, and strategies while participating in • small-sided and lead-up net/wall games (e.g., balloon volleyball, pickleball, handball) and refine selected movement skills, tactics, and strategies while participating in • low-organizational, inventive, and co-operative games (e.g., tag games, relay race, prisoner's base) • small-sided and lead-up target games (e.g., bowling, curling, golf, bocce ball) • small-sided and lead-up striking/fielding games (e.g., long ball, kick ball, softball) • small-sided and lead-up invasion/territorial games (e.g., two-on-two, three-on-three games using skills from games such as soccer, basketball, and soft lacrosse) • alternate environment activities (e.g., hiking, aquatics, skating, snowshoeing, orienteering, cross-country skiing, tobogganing, cycling, tracking)
GCO 1, GCO 3 (*Prince Edward Island Physical Education Safety Guidelines* document.)	**Active Living, Relationships** *Students will be expected to* 4.10 Tactics, Strategies, and Rules Apply tactics, strategies, and rules necessary for safe and inclusive involvement in movement activities, including but not limited to co-operative and competitive lead-up games as well as alternate environment activities, when alone and with others
GCO 2, GCO 3 (*Prince Edward Island Physical Education Safety Guidelines* document.)	**Skilful Movement, Relationships** *Students will be expected to* 4.11 Safety and First Aid Incorporate safe practices (e.g., warm-up, cool-down, safe stretching, protective responses, proper attire, well-maintained equipment) for the prevention of injury, and investigate basic first aid associated with care of illness and injury resulting from participation in movement activities both in and out of physical education class
GCO 3	**Relationships** *Students will be expected to* 4.12 Relationships Create and apply a personal understanding of what it means to be a positive, inclusive team member who makes a commitment towards showing team spirit and the ideals of fair play 4.13 Culture and History Examine and communicate the contributions, both historically and currently, that the Aboriginal people on PEI, the Métis, and other cultures of our province have made to the development of games, sports, and other movement activities

Science

General Curriculum Outcomes

STSE

GCO 1: Students will develop an understanding of the nature of science and technology, of the relationships between science and technology, and of the social and environmental contexts of science and technology. (STSE)

Skills

GCO 2: Students will develop the skills required for scientific and technological inquiry, for solving problems, for communicating scientific ideas and results, for working collaboratively, and for making informed decisions.

Knowledge

GCO 3: Students will construct knowledge and understandings of concepts in life science, physical science, and earth and space science, and apply these understandings to interpret, integrate, and extend their knowledge.

Attitudes

GCO 4: Students will be encouraged to develop attitudes that support the responsible acquisition and application of scientific and technological knowledge to the mutual benefit of self, society, and the environment.

Specific Curriculum Outcomes

Life Science: Habitats

Habitats and Populations
Students will be expected to

- use the terms habitat, population, and community in appropriate contexts (104-6)
- identify questions to investigate about the types of plants and/or animals at a local habitat, and the conditions under which they live (204-1)
- identify various methods for finding answers to questions related to their local habitat, and select one that is appropriate (204-6)
- make observations and collect information related to local habitats and their associated populations of plants and animals (205-5, 302-1)
- identify their own and their families' impact on habitats, and describe how personal actions help conserve habitats (108-6, 108-3)
- compile and display the data collected in the habitat study using tallies, tables, and/or bar graphs (206-2)
- present the procedures and results of their habitat studies, compare their results with those of other class members, recognizing that results may vary, and suggest explanations for these discrepancies (104-4, 206-3)

Collecting Scientific Information using Models of Natural Habitats
Students will be expected to

- construct and/or maintain a model of a natural habitat, and use it to make observations and collect information about organisms in this habitat (205-10, 205-5)
- suggest improvements to the model of the natural habitat to make it more realistic and habitable for organisms (206-6)

Behavioural and Structural Features of Animals That Enable Them to Survive in Their Habitat
Students will be expected to

- compare the external features and behavioural patterns of various animals and relate these features to their ability to meet their basic needs in their natural habitats (302-2, 300-1)
- carry out procedures and ensure a fair test to explore how appearance affects visibility (205-1)
- predict the structural and/or behavioural adaptations needed for an animal to live in a particular habitat, real or imagined (204-3)

General Curriculum Outcomes

Specific Curriculum Outcomes

Structural Features of Plants that Enable Them to Survive in Their Habitat
Students will be expected to

- using appropriate terminology, compare the structural features of plants that enable them to thrive in different kinds of places (300-2, 104-6)
- describe how scientists' knowledge of plant growth has led to agricultural innovations and techniques (106-4)
- describe current investigations into local or regional habitat issues (105-1)

Food Chains
Students will be expected to

- classify organisms according to their role in a food chain and draw a diagram to illustrate the food chain (302-3, 104-6, 206-1)
- predict how the removal of a plant or animal population affects the rest of the community (301-1)
- relate habitat loss to the endangerment or extinction of plants and animals (301-2)

The Impact of Technology on Natural Habitats
Students will be expected to

- identify examples of positive and negative effects of technological developments on natural habitats (108-1)

Physical Science: Light

Optical Devices
Students will be expected to

- describe how knowledge of the properties of light has led to the development of optical devices that extend our ability to observe (106-1, 106-4)
- compare how light interacts with a variety of optical devices (107-1, 303-8)
- construct an optical device that performs a specific function (205-10)
- identify women and men in their community who have careers that deal directly with lenses, mirrors, and prisms (107-10)

Sources of Light
Students will be expected to

- distinguish between objects that emit their own light and those that require an external source to be seen (303-3)
- make observations and collect information during investigations to determine if an object emits its own light, and draw conclusions based on the evidence gathered (205-5, 206-5)
- provide examples of how human-made sources of light have been designed to solve problems in the home and at school (107-4)
- identify positive and negative effects of exposure to light (108-1)
- identify ways of conserving energy through conservative use of home lighting (108-6)

General Curriculum Outcomes

Specific Curriculum Outcomes

Light Radiates from a Source
Students will be expected to

- make observations about how light is dispersed from a variety of light sources (205-5)
- demonstrate that light travels in all directions away from a source (303-2)
- conclude that light travels in a straight line based on evidence gathered through their own research and observation (206-5)

Objects that Absorb, Transmit, and/or Reflect Light
Students will be expected to

- investigate how light interacts with a variety of objects, in order to determine whether the objects cast shadows, allow light to pass, and/or reflect light (303-4)
- classify objects as opaque, transparent, or translucent (206-1)
- predict changes in the location, shape, and relative size of a shadow when an object is placed in different positions and orientations relative to a light source and screen (303-5)
- plan a procedure and make observations to determine changes in a shadow's location, shape, and relative size when an object is placed in different positions and orientations relative to a light source and screen (204-7, 205-5)
- make observations and collect information about the reflective properties of surfaces of different shapes and textures (205-5)

Bending Light
Students will be expected to

- demonstrate and describe how a variety of media can be used to change the direction of light (303-6)
- make observations and collect information about the refractive properties of materials of different shapes (205-5)

Dispersion of Light
Students will be expected to

- demonstrate that white light can be separated into colours, and use the term "dispersion" for this process (303-7, 104-6)
- follow a set of procedures to make and use a colour wheel (205-3)
- communicate and listen to others during investigations with colour wheels (207-1)

Physical Science: Sound

Objects that Make Sounds
Students will be expected to

- identify objects by the sounds they make (303-9)
- describe examples of sound technologies that are used by people to meet their everyday needs (107-1)

General Curriculum Outcomes

Specific Curriculum Outcomes

Sound Vibrations
Students will be expected to

- relate vibrations to sound production (303-10)
- compare how vibrations travel differently through a variety of solids, liquids and through air (303-11)

Pitch, Loudness, and Sound Technology
Students will be expected to

- identify and rephrase questions about ways to change pitch and loudness in a testable form (104-6, 204-1, 204-2)
- state a prediction and hypothesis about the effect a modification will have on the pitch and loudness of the sound produced, based on the pattern of sounds produced (204-3)
- demonstrate and describe how the pitch and loudness of sounds can be modified (301-3)
- use the term "decibel" in descriptions of sound intensity (104-6)
- demonstrate processes for solving technological problems by designing and constructing a device which has the ability to create sounds of variable pitch and loudness (104-1, 205-2)
- evaluate personally constructed musical devices with respect to their ability to vary their pitch and loudness (206-7)
- identify and use a variety of sources and technologies to gather pertinent information about Canadians who have contributed to sound technology (107-12, 205-8)

The Ear, Hearing Loss, and Noise Pollution
Students will be expected to

- describe and illustrate how the human ear is designed to detect vibrations (300-3)
- compare the range of sounds heard by humans to that heard by some animals (300-4)
- describe examples of devices that enhance our abilities to hear and collect sound data, such as hearing aids, sonar, amplifiers, oscilloscopes, and ultrasound (106-1)
- demonstrate processes for investigating the extent of noise pollution in their surroundings, and work with group members to evaluate the processes used in investigating noise pollution (104-1, 207-6)
- identify the positive and negative effects of technological devices that produce loud sounds and identify the need for protection from and prevention of hearing loss (108-1, 206-9)
- describe specific personal actions or products that can help reduce this noise pollution (108-3)
- identify examples of current research related to sound (105-1)

General Curriculum Outcomes

Specific Curriculum Outcomes

Space and Earth Science: Rocks, Minerals, and Erosion

Collecting and Comparing Rocks and Minerals
Students will be expected to

- demonstrate respect for the habitats of animals and the local environment when collecting rocks and/or minerals from their local area (108-3)
- describe the distinction between minerals and rocks (104-6)

Properties of Rocks and Minerals
Students will be expected to

- use appropriate tools to make observations and collect information in order to describe rocks and minerals according to physical properties (204-8, 205-5, 300-6)
- record observations of their rocks and minerals in chart form, and/or using notes in point form (205-7)
- compare different rocks and/or minerals from their local area with those from other places (300-5)
- classify their rocks and minerals according to several properties and create a chart or diagram that show the method of classifying (206-1, 207-2)
- compare their classification schemes of the rocks/minerals to those of others and recognize that results may vary (104-4)

Uses for Rocks and Minerals
Students will be expected to

- describe how rocks and minerals are used (107-1)
- relate the characteristics of rocks and minerals to their uses (300-8)
- using appropriate terms, describe some positive and negative effects of the extraction and/or utilization of rocks and minerals (104-6, 108-1)

Erosion and Weathering
Students will be expected to

- describe effects of wind, water, and ice on the landscape (301-5)
- demonstrate a variety of methods of weathering and erosion (301-6)

Soil Formation and Composition
Students will be expected to

- describe ways in which soil is formed from rocks (301-4)

Records in Rocks
Students will be expected to

- identify and describe rocks that contain records of Earth's history (300-7)

Sudden and Significant Changes in the Land
Students will be expected to

- describe natural phenomena that cause sudden and significant changes to the landscape (301-7)

Social Studies

General Curriculum Outcomes

Citizenship, Power, and Governance

GCO: Students will be expected to demonstrate an understanding of the rights and responsibilities of citizenship and the origins, functions, and sources of power, authority, and governance.

Culture and Diversity

GCO: Students will be expected to demonstrate an understanding of culture, diversity, and world view, recognizing the similarities and differences reflected in various personal, cultural, racial, and ethnic perspectives.

Individuals, Societies, and Economic Decisions

GCO: Students will be expected to demonstrate the ability to make responsible economic decisions as individuals and as members of society.

Interdependence

GCO: Students will be expected to demonstrate an understanding of the interdependent relationship among individuals, societies, and the environment — locally, nationally, and globally — and the implications for a sustainable future.

People, Place, and Environment

GCO: Students will be expected to demonstrate an understanding of the interactions among people, places, and the environment.

Time, Continuity, and Change

GCO: Students will be expected to demonstrate an understanding of the past and how it affects the present and the future.

Specific Curriculum Outcomes

Conceptual Organizer: Explorations

Unit One: Exploration
Students will be expected to

4.1.1 explore the concept of exploration

Unit Two: The Nature of Exploration
Students will be expected to

4.2.1 examine the stories of various explorers of land, ocean, space, and ideas
4.2.2 analyse factors that motivate exploration
4.2.3 evaluate the impact of exploration over time

Unit Three: Exploring Our World
Students will be expected to

4.3.1 examine major physical features of the world
4.3.2 describe the main characteristics of rivers, islands, mountains, and oceans
4.3.3 examine the relationship between humans and the physical environment

Unit Four: Exploring the Landscapes of Canada
Students will be expected to

4.4.1 describe the physical landscape of Canada
4.4.2 examine the human landscape of Canada
4.4.3 demonstrate an understanding of the political landscape of Canada
4.4.4 examine symbols associated with Canada's landscapes

Visual Arts

General Curriculum Outcomes

Strand One: Fundamental Concepts

Students will be expected to develop understanding of the following concepts through participation in a variety of hands-on, open-ended visual arts experiences.

Strand Two: Fundamental Concepts

Students will be expected to apply the creative process to produce a variety of two- and three-dimensional art works, using elements, principles, and techniques of visual arts to communicate feelings, ideas, and understandings.

Specific Curriculum Outcomes

Elements of Art and Design

Students will be expected to

FC4.1 develop an understanding of the elements and principles of art and design in creating and viewing artwork

- line: lines to indicate emotion (e.g., smooth, horizontal lines can give a feeling of peace and harmony)
- shape and form: free-standing forms "in the round" (e.g., Henry Moore's figurative work) and "bas relief sculpture" (e.g., masks); changes in shapes, depending on the angle or point of view from the top, side, bottom)
- space: diminishing perspective in various contexts (e.g., in vertical placement, in diminishing size, and/or in overlapping shapes)
- colour: monochromatic colour scheme
- texture: texture elaboration (e.g., embossing, piercing, pinching, pressing, scoring, scraping)
- value: variations in value to create emphasis (contrast in value)

Principles of Art and Design

Students will develop understanding of all principles of art and design (that is, contrast, repetition and rhythm, variety, emphasis, proportion, balance, unity and harmony, and movement), but the focus in Grade 4 will be on emphasis.

- emphasis: use of colour intensity, contrast

Students will be expected to

CP4.1 create two- and three-dimentional works of art that express feelings and ideas inspired by their interests and experiences
(e.g., a comic strip or a storyboard featuring a space voyage)

CP4.2 demonstrate an understanding of composition, using principles of art and design to create narrative art works or art works on a theme or topic
(e.g., a collaborative mural depicting a historical or an imaginary landscape in which objects and figures placed in the foreground create areas of emphasis, and objects placed in the background show diminishing size)

CP4.3 use a variety of materials, tools, and techniques to respond to design challenges
(e.g., use contour lines of various weights in a charcoal gesture drawing of a person to capture the impression of movement)

CP4.4 use a variety of materials, tools, and techniques to determine solutions to design challenges

- drawing: using a graphite pencil (e.g., primary printers) and depicting the objects from different points of view (e.g., from the front, the back, the side)

General Curriculum Outcomes	Specific Curriculum Outcomes *Students will be expected to*	
		- mixed media: make a collage to depict a dream, using cut and torn paper, tissue paper, and found objects in contrasting shapes with a focus on positive and negative space - painting: use tempera paint and a range of monochromatic colour values to represent the emotional state of a character at a critcal moment in a story that they have written or read - printmaking: use low-relief found objects (e.g., lace, textured leaves, and tin foil) to make a collograph in which texture and shape are used to create the composition, and embellish the final inked print with oil pastel drawing - sculpture: make a clay or papier-mâché mask featuring exaggeration for dramatic effect and textures made by embossing, piercing, pinching, pressing, and/or scraping
Strand Three: Reflecting, Responding, and Analysing Students will be expected to apply the critical analysis process to communicate feelings, ideas, and understandings in response to a variety of art works and art experiences.	RR4.1	express personal feelings and ideas about art experiences and images (e.g., images that promote business, events, or festivals)
	RR4.2	demonstrate an awareness of a variety of art forms, styles, and traditions, and describe how they reflect the diverse cultures, times, and places in which they were made (e.g., contemporary and historical oil paintings in an art gallery)
	RR4.3	demonstrate an awareness of the meaning of signs and symbols encountered in their daily life lives and in works of art (e.g., symbols representing luck; fonts typically used in marketing; heraldic symbols; Aboriginal totems around the world; Egyptian hieroglyphics)
	RR4.4	identify and document their strengths, their interests, and areas for improvement as creators and viewers of art (e.g., review notes and sketches they have made during a visit to a public gallery, and summarize what tends to interest them when they look at art)
Strand Four: Exploring Forms and Cultural Contexts Students will be expected to demonstrate an understanding of a variety of art forms, styles and techiques from the past and present, and their social and/or community contexts.	EC4.1	identify and describe some of the ways in which art forms and styles reflect the beliefs and traditions of a variety of communities, times, and places (e.g., differences in styles among different artists can be associated with a specific reason, intent, or motivation)
	EC4.2	demonstrate an awareness of a variety of works of art and artistic traditions from diverse communities, times, and places (e.g., artists contribute to Canada's economy by providing both goods and services)